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Inventor(s)	Hall; Taylor James

Saw guide assembly, saw system, and methods

Abstract

A shuttle assembly for use with a saw guide track is provided. The shuttle assembly includes a base plate to which a saw is mounted. An alignment spacer extends laterally outward from the base plate and provides an abutment for locating a saw blade of the saw while the saw is being secured to the shuttle assembly.

Inventors:	Hall; Taylor James (Palatine, IL)
Applicant:	BAM IP Holding LLC (Saint Charles, IL)
Family ID:	1000008820140
Assignee:	BAM IP Holding LLC (Saint Charles, IL)
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Attorney, Agent or Firm: Reinhart Boerner Van Deuren P.C.

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This patent application is a continuation of U.S. patent application Ser. No. 17/196,757, filed Mar. 9, 2021, which is now, the entire teachings and disclosure of which are incorporated herein by reference thereto.

(1) This invention generally relates to track saws.

(2) Track saw systems use a track and shuttle assembly for guiding a saw along a predetermined path. This can provide for highly accurate cutting of a workpiece. Some track saw systems have dedicated saws that are not used for other activities. Some track saw systems convert a saw used for other activities into a track saw. For example, a handheld circular saw may be used in a track saw system to gain improved cutting accuracy, e.g. straighter cuts, than using the circular saw free-handed.

(4) In many systems, the shoe of the circular saw (i.e. the portion of the circular saw that rests on top of a workpiece) is abutted against an abutment of the shuttle assembly to orient the saw relative to the shuttle. Unfortunately, the position of the shoe in many circular saws has a low tolerance relative to the position of the saw blade, even prior to use. Further, circular saws are often dropped or can become otherwise damaged when they are on a worksite. This can further affect the orientation of the saw blade relative to the shoe.

BRIEF SUMMARY OF THE INVENTION

(7) In one example, the alignment spacer is also movable relative to the base plate once a saw has been attached to disengage the saw blade from the alignment spacer.

- (8) In one example, the alignment spacer is removably attachable to the base plate.
- (9) In one example, the base plate has a groove formed in the bottom side. The alignment spacer includes an attachment feature in the form of a mounting head matable with the groove. The mounting head is laterally spaced from the saw blade abutment.
- (10) In one example, the alignment spacer includes an attachment feature in the form of threaded holes or threaded studs that can cooperate with corresponding attachment features of the base plate.
- (11) In one example, the mounting head is elongated parallel to the guide axis and prevents pivotal movement of the alignment spacer relative to the base plate along any axis that is non-parallel to the guide axis.
- (12) In one example, an attachment arrangement between the alignment spacer and the base plate fixes the lateral spacing of the saw blade abutment and maintains the saw blade abutment parallel to the guide axis.
- (13) In one example, the alignment spacer includes: a main body portion; a downward depending flange portion extending downward from the main body and defining the saw blade abutment; and an attachment feature for affixing the alignment spacer to the base plate. The main body portion extends between and interconnects the downward depending flange and the attachment feature.
- (14) In one example, the alignment spacer further includes an upward extending flange extending upward from the main body portion. The upward extending flange is laterally between the downward depending flange and the attachment feature. The upward extending flange is spaced from the attachment feature and the downwardly depending flange. The upward extending flange abuts the bottom side of the base plate when the alignment spacer is attached to the base plate.
- (15) In one example, the attachment feature is an upward extending projection that extends upward beyond a top surface of the main body portion.
- (16) In one example, the upward extending flange, downward depending flange, attachment feature and main body portion are all formed from a single continuous piece of material.
- (17) In one example, an axial length of the saw blade abutment is less than an axial length of the base plate between the first and second ends.
- (18) In one example, the saw attachment arrangement is at least one clamp that presses a shoe of the saw against the top of the base plate to secure the saw to the base plate.
- (19) In an example, a saw guide system includes a shuttle assembly as outlined above and a saw guide track configured to mate with the shuttle. The shuttle is slidable relative to the saw guide track along the predetermined path.
- (20) In one example, the saw guide track has a length parallel to the guide axis that is greater than a length of the base plate between the first and second ends.
- (21) In one example, a saw system includes a saw guide system as outlined above and a saw. The saw has a saw blade and a shoe. The saw blade extends downward beyond a bottom side of the shoe. The shoe rests on the top of the base plate when the saw attachment arrangement secures the saw to the shuttle assembly.
- (22) In an example, a method of mounting a saw to a shuttle assembly as outlined above is provided. The method includes positioning the alignment spacer such that the saw blade abutment is spaced laterally outward from the lateral side of the base plate; a butting a saw blade of the saw against the saw blade abutment of the alignment spacer; securing the saw to the base plate using the saw attachment arrangement; and moving the alignment spacer relative to the base plate to disengage the saw blade from the alignment spacer.
- (23) Other aspects, objectives and advantages of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings incorporated in and forming a part of the specification illustrate several aspects of the present invention and, together with the description, serve to explain the principles of the invention. In the drawings:
- (2) FIG. 1 is a perspective illustration of a saw system according to an example of the disclosure;
- (3) FIG. 2 is a perspective view of the saw guide system of the saw system of FIG. 1 illustrating the bottom of the shuttle assembly and a top of the saw guide track;
- (4) FIG. 3 is an exploded illustration of the saw and shuttle assembly of the system of FIG. 1;
- (5) FIG. 4 is an end view of the saw and shuttle assembly of FIG. 3 in an assembled configuration;
- (6) FIG. 5 is an end view illustration of FIG. 4 with the saw and shuttle assembly disconnected from one another;
- (7) FIG. 6 is a bottom view of the saw and shuttle assembly illustrating the alignment spacer in place for mounting the saw to the shuttle assembly;
- (8) FIGS. 7 and 8 illustrate the shuttle base plate and alignment spacer of the shuttle assembly of FIGS. 5 and 6; and
- (9) FIG. 9 is a top view illustration of the saw attached to the shuttle assembly.
- (10) While the invention will be described in connection with certain preferred embodiments, there is no intent to limit it to those embodiments. On the contrary, the intent is to cover all alternatives, modifications and equivalents as included within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE INVENTION

- (11) FIG. 1 illustrates a saw system **100** according to an example of the disclosure. The saw system **100** is configured to provide accurate cutting of workpiece **102** by saw **104**.
- (12) The saw system **100** generally includes saw **104**, saw guide track **106** (also referred to herein simply as track **106**), and a shuttle assembly **108**. The saw guide track **106** and shuttle **108** may be considered a “saw guide system **109**.” In this example, the saw **104** is affixed to the shuttle assembly **108** and the shuttle assembly **108** is slidable (illustrated by arrow **112**) relative to the track **106** along a predetermined path. Typically, the predetermined path is a linear path illustrated by axis **110**.
- (13) The shuttle assembly **108** and saw guide track **106** cooperate with one another to prevent substantially any lateral movement relative to axis **110** or angular movement therebetween such that saw **104** is precisely guided along the predetermined path.
- (14) The track **106** may be formed by one or more sections. The track **106** in this example is formed from a pair of interconnected sections.
- (15) With additional reference to FIG. 2, the track **106** has a bottom **114** that rests on the workpiece to be cut and a top **116** upon which the shuttle assembly **108** rests.
- (16) The shuttle assembly **108** includes a guide arrangement **118** that mates with a corresponding guide member **120** of the track **106**. In this example, the guide arrangement **118** includes a groove **122** that receives the guide member **120**. The guide member **120** is in the form of an upstanding wall that is sized to the width of the groove **122** to eliminate clearance between the guide arrangement **118** and the guide member **120** and promote an accurate cut as the shuttle assembly **108** slides along track **106**.
- (17) The groove **122** defines a guide axis **124** that is parallel to path **110**. Again, the cooperation of guide member **120** and guide arrangement **118** prevents lateral movement, e.g. perpendicular to axes **124**, **110**, of the shuttle assembly **108** relative to track **106**.
- (18) In this example, the track **106** is formed from metal, such as extruded aluminum, and the guide arrangement **118** is formed from plastic, preferably a plastic that has low frictional characteristics relative to track **106**.
- (19) With reference to FIG. 3, the shuttle assembly **108** includes a base plate **130** and a saw attachment arrangement as well as guide arrangement **118**. The guide arrangement **118** in this example is provided by a separate body secured to the base plate **130**. However, in other examples, the guide

arrangement **118** could be directly formed by the base plate **130**.

(20) The saw attachment arrangement is provided by a plurality of clamps **132**. The clamps **132** provide axial loading to the shoe **134** of the saw **104** to press the bottom **136** of the shoe **134** against a top **138** of the base plate **130**.

(21) When the saw guide system **109** isn't used with saw **104**, the bottom **136** of shoe **134** will rest on a top surface of a workpiece when the user cuts the workpiece, such as when the user simply uses sight to align the saw **104** with a desired cutting path.

(22) The saw attachment arrangement (e.g. clamps **132**) of this example allow for adjustably positioning the saw **104** relative to the base plate **130**. The saw attachment arrangement also allows for saws with different sized or shaped shoes **134** to be easily mounted to the base plate **130**.

(23) In the past, a user would typically abut one of the ends **140**, **142** or side **144** of the shoe **134** against an abutment of the shuttle assembly **108** to attempt to properly align the saw **104**, and particularly saw blade **146** thereof, relative to the shuttle assembly **108**.

(24) Unfortunately, the relative orientation of the saw blade **146** relative to the shoe **134** is typically not held to high tolerances. Further, the relative orientation of the saw blade **146** to the shoe **134** can change over time due to wear or damage such as if the saw were dropped. Thus, ends **140**, **142** are often not perpendicular to the cutting plane of saw blade **146** and side **144** is often not parallel to the cutting plane of saw blade **146**. As such, using the shoe **134** as a reference point for mounting the saw **104** to the shuttle assembly **108** has been determined to provide a source of inaccuracy when using a saw guide system of the type similar to saw guide system **109**.

(25) With additional reference to FIGS. 4-6, an alignment spacer **150** is provided to rectify this issue. The alignment spacer **150** includes a saw blade abutment **152** against which an inner side **154** of the saw blade **146** abuts while the user secures the saw **104** to the shuttle assembly **108**.

(26) Here, the saw blade **146** is directly used to locate and orient the saw blade **146** relative to the base plate **130** of the shuttle assembly **108**.

(27) Once the saw **104** is secured to the shuttle assembly **108**, the alignment spacer **150** can be removed or otherwise moved out of engagement with the saw blade **146**.

(28) The alignment spacer **150** is configured to laterally offset the saw blade abutment **152** from lateral side **156** of the base plate **130** of the shuttle assembly **108**. Thus, when the saw **104** is attached to the shuttle assembly **108**, the saw **104** will extend laterally outward beyond lateral side **156** such that the saw blade **146** can extend downward below bottom **158** of the base plate **130**.

(29) In this example, the alignment spacer **150** is removably attachable to the base plate **130**. It is contemplated that in other examples, the alignment spacer **150** could simply slide laterally inward rather than be removed when the shuttle assembly **108** is used with track **106**.

(30) However, regardless of how the alignment spacer **150** is moved out of the way for use of the saw **104**, the alignment spacer needs to be laterally fixable relative to the base plate **130** during mounting of the saw **104**.

(31) In this example, the saw blade abutment **152** defines an abutment plane that is parallel to the guide axis **124** of the guide arrangement **118**. This properly orients the saw blade **146** of the saw **104** relative to the guide axis **124**.

(32) With additional reference to FIGS. 7 and 8, in this example, the alignment spacer **150** includes an attachment feature in the form of a mounting head **160** to orient and secure the alignment spacer **150** to the base plate **130**. The mounting head **160** is matable with a groove **162** formed in an underside of base plate **130**.

(33) Preferably, the mounting head **160** is sufficiently elongated relative to the size and shape of groove **162** such that the mounting head **160** does not move laterally relative to guide axis **124** within groove **162** nor does mounting head **160** twist angularly about an axis **166** perpendicular to guide axis **124** when lateral inward directed pressure, illustrated by arrow **168** in FIG. 8, is applied to the alignment spacer **150**.

(34) With reference to FIG. 8, the alignment spacer **150** includes a main body portion **170**, a downward depending flange portion **172** that defines saw blade abutment **152**, and an attachment feature in the form of mounting head **160**. Further, in this example, a support rib **174** formed by an upward extending flange is positioned laterally between flange portion **172** and mounting head **160**. The support rib **174** presses on an underside of base plate **130** for added stability.

(35) The downward depending flange portion **172** can be hooked over an edge of a workpiece or a workbench while mounting the saw **104** to the shuttle assembly **108**. This further stabilizes the shuttle assembly **108** during mounting and allows the saw blade to extend downward below the base plate **130**.

(36) In the illustrated example, the downward depending flange portion **172**, the main body portion **170**, the downward depending flange portion **172** and the support rib **174** are formed from a continuous piece of material, such as being molded as a single body, such as being extruded, or being machined from a single piece of material.

(37) FIG. 9 illustrates the saw **104** attached to the shuttle assembly **108** using clamps **132** pressed axially against a top of the shoe **134**.

(38) By using the clamps **132**, the position of the saw **104** can be adjusted relative to base plate **130** angularly such as illustrated by arrow **180** about axis **182** or linearly as illustrated by arrow **184**.

(39) In this example, the length of the base plate **130** is less than the length of the track **106** of the segments of the track **106**. Further, the length of the alignment spacer **150** is less than the length of the base plate **130**.

(40) A method of mounting a saw **104** to the shuttle assembly includes positioning the alignment spacer **150** such that it extends laterally outward beyond side **156** of the base plate **130**. Abutting a saw blade **146** against the saw blade abutment **152** of the alignment spacer **150** and then securing the saw **104** to the base plate **130**. Thereafter, the alignment spacer **150** is disengaged from the saw blade **146**.

(41) All references, including publications, patent applications, and patents cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

(42) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) is to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

(43) Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

Claims

1. A saw guide assembly for guiding a saw having a saw blade along a path, the saw guide including: a base plate having opposed top and bottom sides, opposed first and second ends, and a lateral side extending between the first and second ends; a saw attachment arrangement for adjustably attaching the saw to the top side of the base plate with the saw extending laterally outward beyond the lateral side such that the saw blade can extend

downward beyond the bottom side of the base plate; and an alignment spacer having a saw blade abutment, the alignment spacer being engageable with the base plate such that the saw blade abutment against which the saw blade abuts while the user secures the saw to the base plate, is spaced laterally outward from the lateral side and such that the saw blade abutment is fixed along a second axis being perpendicular to the path when the alignment spacer is engaged with the base plate.

2. The saw guide assembly of claim 1, wherein the alignment spacer is removably attachable to the base plate.

3. The saw guide assembly of claim 1, wherein: the base plate has a groove formed in the bottom side; the alignment spacer includes an attachment feature in the form of a mounting head matable with the groove, the mounting head being laterally spaced from the saw blade abutment; and mating of the mounting head with the groove prevents the alignment spacer from moving laterally relative to the base plate along the second axis.

4. The saw guide assembly of claim 3, wherein the mounting head is elongated parallel to the path and prevents pivotal movement of the alignment spacer relative to the base plate along any axis that is non-parallel to the path.

5. The saw guide assembly of claim 1, further comprising an attachment arrangement between the alignment spacer and the base plate fixing the lateral spacing of the saw blade abutment and maintaining the saw blade abutment parallel to the path.

6. The saw guide assembly of claim 5, wherein the alignment spacer includes: a main body portion; a downward depending flange portion extending downward from the main body and defining the saw blade abutment; and an attachment feature for laterally affixing the alignment spacer to the base plate, the main body portion extending between and interconnecting the downward depending flange and the attachment feature.

7. The saw guide assembly of claim 6, wherein the alignment spacer further includes an upward extending flange extending upward from the main body portion, the upward extending flange being laterally between the downward depending flange and the attachment feature and being laterally spaced from the attachment feature and the downwardly depending flange, the upward extending flange abutting the bottom side of the base plate when the alignment spacer is attached to the base plate.

8. The saw guide assembly of claim 6, wherein the attachment feature is an upward extending projection that extends upward beyond a top surface of the main body portion.

9. The saw guide assembly of claim 7, wherein the upward extending flange, downward depending flange, attachment feature and main body portion are all formed from a single continuous piece of material.

10. The saw guide assembly of claim 1, wherein an axial length of the saw blade abutment is less than an axial length of the base plate between the first and second ends.

11. The saw guide assembly of claim 1, wherein the saw attachment arrangement is at least one clamp that presses a shoe of the saw against the top of the base plate to secure the saw to the base plate.

12. A saw system comprising: the saw guide assembly of claim 1; and a saw having a saw blade and a shoe, the saw blade extending downward beyond a bottom side of the shoe, the shoe resting on the top of the base plate when the saw attachment arrangement secures the saw to the saw guide.

13. A method of mounting a saw to a saw guide assembly of claim 1, the method comprising: positioning the alignment spacer such that the saw blade abutment is spaced laterally outward from the lateral side of the base plate; abutting a saw blade of the saw against the saw blade abutment of the alignment spacer; securing the saw to the base plate using the saw attachment arrangement; and moving the alignment spacer relative to the base plate to disengage the saw blade from the alignment spacer.

14. A saw system comprising: the saw guide assembly of claim 1; a saw mounted to the base plate of the saw guide, the saw having a saw blade extending downward beyond the bottom side of the base plate of the saw guide; and the alignment spacer removably attached to the base plate, the saw blade abutment abutting against the saw blade.

15. The saw system of claim 14, wherein the alignment spacer is removable from the base plate when the saw is secured to the saw guide assembly.

16. The method of claim 13, wherein the step of moving the alignment spacer includes removing the alignment spacer from the base plate prior to cutting with the saw.

17. The saw system of claim 14, wherein the saw includes a shoe, the shoe of the saw mounting the saw to the base plate of the saw guide assembly, the saw blade extending downward beyond a bottom side of the shoe.

18. The saw guide assembly of claim 1, wherein the alignment spacer is engageable with the base plate such that the saw blade abutment is fixed along the second axis being perpendicular to the path when the alignment spacer is engaged with the base plate and the saw is not fixed relative to the base plate.

19. The saw system of claim 14, further including: a guide arrangement configured to laterally locate the base plate relative to the path and inhibit lateral movement of the saw guide assembly relative to a work piece when cutting a workpiece with the saw attached to the base plate as the base plate and saw travel along the path as the base plate.
